Received: 17 November 2021 | Accepted: 2 December 2021

DOI: 10.1002/jpln.202100385

GAME CHANGER

Perspectives from the Fritz-Scheffer Awardee 2020—The mutual interactions between roots and soil structure and how these affect rhizosphere processes[#]

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[#] This article has been edited by Hermann Jungkunst.

Abstract

Roots growing into soil interact with soil structure in numerous ways. They can grow into the soil matrix and leave elongated macropores after decomposition, that is, biopores. Conversely, the soil may already have a large and connected macropore system through which the roots can expand, and thus reach deeper soil layers. Both, the formation of new biopores or the reuse and occupation of existing macropore systems are expected to affect major soil processes like water and gas flow through the soil profile, as well as water flow toward the root. Despite the increasing research interest in the limitation of root growth by soil structure, as well as the modification of soil structure by roots, the mutual interactions between the two are largely overlooked. This study highlights new methodological developments which enable describing interactions between roots and soil structure with X-ray computed microtomography. It further shows how roots affect the pore system and can create a massive biopore system in less than a decade. After this, it is evaluated how the mutual interaction of roots and structure determines the physical properties of the rhizosphere. It is outlined that this has implications for major rhizosphere processes. Thus, it is emphasized, that the role of structure needs to be considered in future experiments with plants, in particular if extrapolation of results from laboratory experiment with sieved, homogenized substrates to field conditions with well-established soil structure is intended. Lastly, in this study, research gaps are outlined remaining in respect to the dynamics of biopore creation and destruction and their consequences for processes in rhizospheres like carbon storage. These reveal the need for novel research approaches that consider the mutual interactions of root growth and soil structure.

KEYWORDS
biopores, macropores, rhizosphere, structure formation, X-ray CT

1 | INTRODUCTION

Growing roots interact with soil and its structure across a range of spatial and temporal scales, and thus adapt to their local environment (Downie et al., 2015). Once they die and decay, they leave behind pores or channels known as biopores (Lucas et al., 2019b; Yunusa & Newton, 2003). In addition to roots, various burrowing organisms, such as earthworms and the fungal hyphae, form biopores. Thus, biopore size ranges from a few micrometers to millimeters in diameter (Yunusa & Newton, 2003). Biopores are cylindrical in shape and can extend over the entire soil profile, showing low tortuosity and high vertical continuity (Kautz, 2015). The biopore properties influence soil water infiltration and preferential flow phenomena (Jarvis, 2007; Koestel & Larsbo, 2014; Naveed et al., 2013; Rasse et al., 2000; Wuest, 2001). Particularly in dense subsoils, root growth seems to benefit from an existing biopore network (Gao et al., 2016). This shows that there can be a close relationship between biopore (network) legacy, new root growth and root water extraction (Stirzaker et al., 1996). For the analysis of biopores, however, there are a number of methodological approaches, which are often limited by the spatial resolution (Athmann et al., 2013; Han et al., 2015; Kautz et al., 2013; Stirzaker et al., 1996; White & Kirkegaard, 2010). Measurements along a profile wall, for example, allow a high spatial resolution, but are only two (2D)-dimensional, while endoscopy only allows the investigation of relatively large pores in the millimeter range. Thus, research in recent decades has mainly focused on biopores >1 mm, and quantification of the temporal dynamics of biopore networks as a result of root growth is missing (Kautz, 2015). Biopores <1 mm, however, represent the majority of all biopores and are important for the connectivity of the macropore system (Lucas et al., 2021).

Roots influence soil structure beyond creating biopores, they also affect the physical properties of their surroundings, that is, the rhizosphere (Aravena et al., 2011; Bruand et al., 1996; Dexter, 1987). Roots pushing through the soil may lead to a compaction of the rhizosphere, depending on the connectivity of the macropore system they encounter (Lucas et al., 2019a). In addition, soil structure is also indirectly influenced by roots through water uptake or the release of carbon. The uptake of root water can locally lead to changes in swelling and shrinking dynamics (Carminati et al., 2013; Helliwell et al., 2019), while the release of organic compounds may stabilize macropore structures (Pierret et al., 2011). As a consequence, at multiple spatial scales (e.g., aggregates, pore network, horizons and profiles), unique structural features emerge through the action of plant roots (Vetterlein et al., 2020). Therefore, structures of soils provide a time-related “memory” of the past (Lin, 2010). However, describing and disentangling these structures, and, above all, to assign them to the processes by which they were created in a chronological order, is a methodological challenge in naturally developed soils.

In general, there are no universal methods to define and characterize soil structure, which describes the spatial arrangement of solids and voids and its development (Dıaz-Zorita et al., 2002; Rabot et al., 2018). It can be described both from the solid phase perspective and from the pore space perspective, as these are complementary aspects (Rabot et al., 2018). Bulk density or aggregate stability and aggregate size distribution are often used as an indicator for soil structure from the solid phase perspective (Amézketa, 1999; Felde et al., 2021; Schweizer et al., 2019). However, the description of aggregates may not allow to link soil structure to all soil functions and processes, which is especially true for biological activity and flow and transport phenomena (Rabot et al., 2018; Vogel et al., 2021). For example, water flow and gas diffusion are both directly affected by the pore space and its size distribution (Koestel et al., 2018; Naveed et al., 2013). Discretizing the pore space into different sizes, that is, by a pore size distribution (PSD), is useful; however, it is also a conceptualization of the true continuous pore space, which does not consist out of a discrete number of round objects (Nimmo, 2013). Describing pore shape and continuity not only enables deriving which abiotic and biotic processes contributed to their formation but also to infer their relevance for system properties (Lucas et al., 2021). In this context, macroporosity and its connectivity are important determinants of soil structure (Rabot et al., 2018). The description of biopores and their size distribution itself can further help to reveal the direct influences of plants and their roots over time (Lucas et al., 2019b, 2021).

X-ray computed microtomography (X-ray µCT) is a noninvasive imaging method, which has been increasingly used in recent years to describe pore systems and their dynamics. Especially, recent developments in image analysis tools make it possible to derive all of the above-mentioned parameters, that is, connectivity and size distribution of pores and biopores as well as rhizosphere physical properties at a high spatial resolution. Thus, it became a suitable tool for the description of the mutual interactions of plant roots and soil structure (Kravchenko & Guber, 2021; Vogel et al., 2021).

The objectives of this study are (1) to address methodological challenges and recent developments in the description of soil structure and plant roots with X-ray µCT and (2) to highlight the direct influences of roots on soil structure and vice versa. Further, (3) it is evaluated how two-way interaction of roots and structure determines the physical properties of the rhizosphere and how major rhizosphere processes such as water flow are affected. Lastly, (4) critical knowledge gaps are identified and concluded by future developments needed.

2 | X-RAY CT AS A TOOL—POSSIBILITIES AND DRAWBACKS

X-ray µCT has the great advantage of noninvasively characterizing both, the architecture of the root system and the pore architecture. However, the output of µCT is depending highly on image quality (e.g., resolution and noise) and image processing as well as analysis workflows. Especially, objects close to the image resolution cannot be captured in a representative way due to imaging artefacts (Vogel et al., 2010). Thus, fine roots as well as their resulting biopores are hard to be segmented properly, as objects smaller five voxels are hard to quantify during image processing and analysis. However, developments in image analysis enable how to classify roots with diameters near image

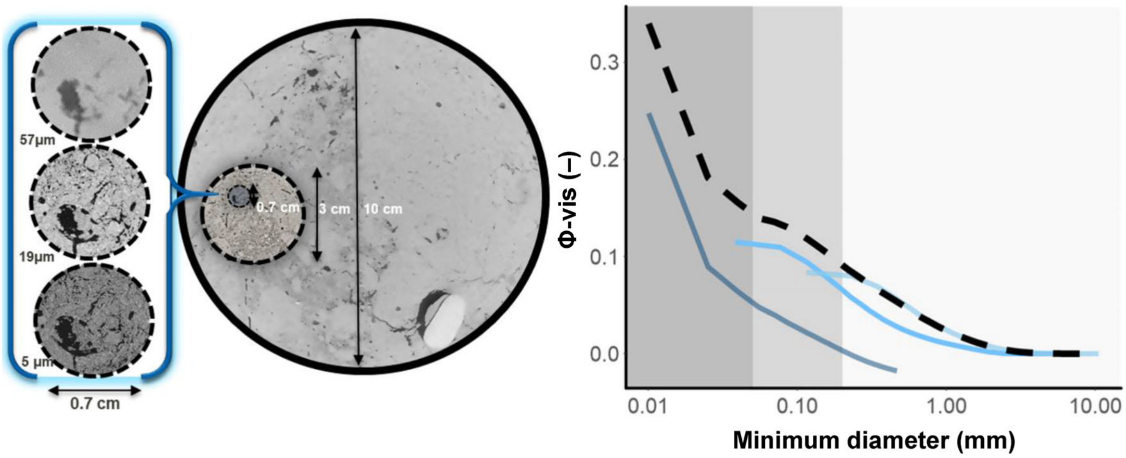

FIGURE 1 Trade-off between sample size and resolution. Left, 2D slices of 10 cm, 3 cm, and 0.7 cm samples, which show the appearance of small image features visible in small samples. Right, respective pore size distributions of the three image sizes in blue (darker color for decreasing sample size) and their joint distribution in black (Lucas et al., 2020)

resolution (Flavel et al., 2017; Gao et al., 2019; Phalempin et al., 2021a). Now it is possible to visualize the associated plant–soil interactions at biologically relevant (sub-) millimeter scales up to near image resolution of µCT in soils.

3 | OVERCOMING SCALE DEPENDENCIES—DESCRIBING STRUCTURE ACROSS SCALES

There is a strong trade-off between sample size and resolution during µCT scanning, that is, small sample sizes (<1 cm) are needed to derive small resolutions (<0.01 mm). These, however, are not representative capturing large components of soil structures (larger biopores and cracks). A possibility to minimize this trade-off is to combine different sample sizes (Vogel et al., 2010). Lucas et al. (2019b) was the first study to combine three different sample sizes to describe the pore size distribution representative down to a resolution of 0.005 mm (Figure 1). Such a joint pore size distribution captures a wide range of pore types like large biopores created by earthworms and tap roots (>1 mm), but also biopores created by fine roots (<1 mm) and small pores like pores within biopore walls as well as packing pores (<0.1 mm).

Many pore metrics such as pore connectivity may be scale-dependent (Lucas et al., 2021). Several mathematical approaches can be used to characterize 3D connectivity in porous media (Renard & Allard, 2013). Two of them are the Euler number, χ and the connection probability of two random points within the pore system, that is, the Γ-indicator. The Euler number is a topological measure from integral geometry, and small-scale features contribute to χ in the same way as large-scale features. However, small-scale features are typically much more abundant, which makes χ highly sensitive to small pores. In contrast, the Γ-indicator is based on pore cluster size distribution and thus is sensitive to large macropores. Lucas et al. (2021) investigated the scale dependency of these two connectivity measures in detail. Their results showed that unlike χ, Γ provides highly biased information in small samples as long-distance connections (created by biopores) are destroyed by subsampling. Hence, the authors created a joint-Γ-curve, like the above-mentioned joint-PSD to reduce this artefact. The authors further showed that the joint evaluation of the two connectivity metrics can be used to unravel different pore types with χ and to identify the contribution of different pore types to total pore connectivity with the Γ–indicator (Lucas et al., 2021).

4 | ROOT AND BIOPORES—SEGMENTATION AND DESCRIPTION

A crucial step during µCT analysis, which describes the interaction of roots and pore structure is the segmentation of the root system as well as the biopore system. This implies the separation of all image voxels into classes, in this case root voxels, biopores and soil matrix (including pores filled with air or water).

One of the easiest segmentation methods is based on the attenuation density, that is, by applying simple thresholds for gray values. This fast segmentation based on the histogram was often applied in early µCT studies (Heeraman et al., 1997; Lontoc-Roy et al., 2005, 2006; Pierret et al., 1999). However, simple thresholding causes major drawbacks, as roots show a similar attenuation as water or other organic material. To overcome these obstacles, more complex imaging approaches were developed and used in recent years. These make use of features inherent in root systems such as connectivity or their cylindrical shape. Examples are Region Growing approaches, that is, adaptive local thresholding methods, which are based on the connectivity of the root system and, for example, implemented in the widely used software VGStudio (Atkinson et al., 2020; Blaser et al., 2018; Flavel et al., 2012; Helliwell et al., 2017, 2019; Koebernick et al., 2014; Tracy et al., 2013) or developed software such as RootViz3D (http://www.rootviz3d.org/) and RooTrak (Mairhofer et al., 2012) as well as Root1 (Schindelin et al., 2012).

A recently developed root segmentation tool by Gao et al. (2019) is called Rootine. It was further improved by Phalempin et al. (2021a). Like Root1, it is a macro for the software Fiji. However, different to other root segmentation methods, it does not directly rely on gray values, that is, attenuation or connectivity, but is shape based. In short, it finds roots by their tubular structure. An important advantage of Rootine is that it does not necessarily rely on the full connectivity of the root system. Thus, also roots in subsamples can be segmented, which are not fully connected any more (Lucas et al., 2019a). In addition, the same approach can be used to segment biopores, that is, cylindrical shaped pores, which have the shape, but not the attenuation value of roots (Lucas et al., 2019b).

The segmentation of images allows to describe the root and biopore system with respect to their spatial extension (volume, length density). In addition, a segmentation enables to describe other traits, such distribution of distance between roots in 3D, which can be related to exploration efficiencies of roots. In addition, these distances can be combined with other parameters, such as porosities and pore sizes. This allows for the description of physical properties within the rhizosphere and within biopores walls in comparison to the bulk soil (Lucas et al., 2019a). This can be linked to rhizosphere processes such as water flow to the roots (e.g., Landl et al., 2021).

To link structural properties with biogeochemical processes, information derived from X-ray µCT needs to be combined with other imaging methods. The combination of different biogeochemical imaging methods, designated as correlative microscopy or correlative imaging, is increasingly used in life sciences (Caplan et al., 2011; Guyader et al., 2018; Handschuh et al., 2013). Schlüter et al. (2018) highlighted that a combination of different 2D techniques for chemical and microbial imaging, such as scanning electron microscope-energy dispersive using X-ray analysis, nanoscale secondary ion mass spectrometry as well as fluorescence microscopy with X-ray µCT, can be used to reveal specific soil microenvironments, and thus biogeochemical and physical processes in structured soil. Lucas et al. (2020) developed a novel correlative analysis of 2D imaging visible light near-infrared spectroscopy and 3D X-ray µCT to achieve detailed understanding of the soil structure and its relation to soil organic matter (SOM) accumulation (Lucas et al., 2020). The method combination allowed observing SOM spatial distribution as a function of distance to features derived from X-ray µCT such biopores.

5 | ROOTS AND SOIL STRUCTURE

The tip of a growing root can either elongate into the dense matrix by overcoming its penetration resistance or use existing macropores (of its size or larger) to grow into soil (Figure 2). The first leads to a reorganization of the pore space and leaves behind a biopore after decay. In contrast, roots growing into existing pores do not change the pore space actively but block the pore they encounter at least partially until they are degraded (Gish & Jury, 1983; Scanlan, 2009). While roots grow into the soil, the physical, chemical and biological properties in their surroundings, that is, in the rhizosphere, are also being modified (Hinsinger et al., 2009; Young, 1998). It was generally assumed that the rhizosphere of roots is compacted compared to the bulk soil, depending on the root diameter (Bruand et al., 1996; Dexter, 1987). This seemed to be reasonable, as roots are typically larger than most pores and thus roots would lead to a compaction of their surrounding as they push aside soil particles (Dexter, 1987). However, in recent years, the use of techniques such as µCT enabled to visualize porosity gradients around roots (Feeney et al., 2006; Helliwell et al., 2017; Vollsnes et al., 2010). These have revealed both increased and decreased porosity in the rhizosphere (Rabbi, Tighe, Flavel et al., 2018; Rabbi, Tighe, Knox et al., 2018; Helliwell et al., 2017, 2019; Koebernick et al., 2019; Vollsnes et al., 2010). The main reason for the latter is that roots grow predominately along the path of least resistance and thus into macropores, which leads to highly porous rhizospheres in well-structured soils (Lucas et al., 2019a).

In addition, different studies have also described an increased porosity in the immediate vicinity of the epidermis of a root followed by a compacted zone (Lucas et al., 2019a; Koebernick et al., 2019; Phalempin et al., 2021a). This was addressed to a “surface/wall effect,” that is, the effect of the packing geometry of spherical soil particles close to the flat root surface (Koebernick et al., 2019).

Soils and their structures, however, are extremely heterogenous and dynamic. Thus, growing roots undergo all kinds of potential interactions with their surroundings, that is, at one point in space and time they may grow through a macropore and directly penetrating back into the soil matrix. Elsewhere, they may follow an existing macropore, such as a crack or a biopore, over longer distances (Jin et al., 2013; White & Kirkegaard, 2010). In general, however, plant roots respond to soil structure by growing into more favorable environments to avoid stressful soil environments (Bengough et al., 2011; Jin et al., 2017). Thus, a biopore created by a root, can later be favorable for a root of the same plant or a following crop to elongate into soil and bypass zones of high resistance (Lucas et al., 2019a; Jin et al., 2013; White & Kirkegaard, 2010). This underlines the complex nature of the feedback chain of roots and soil structure.

The sizes of biopores found in soils can be linked to the plants or more specific their root types used within a crop rotation. Already in 1983, Tippkötter (1983) could link the connected network of biopores in a loess soil, having a maximum between 0.15 and 0.25 mm, to the morphology of cereal roots. This was achieved by analyzing polished sections of impregnated soil samples (Tippkötter, 1983). In a space for time chronosequence of a recultivation area (open-cast lignite mine Garzweiler; details in Pihlap et al., 2019), Lucas et al. (2019b) were able to follow the changes in soil structure induced by plants from the beginning using X-ray µCT (Figure 3). This methodological approach enabled to follow the development of biopores in 3D down to a diameter of 0.038 mm and revealed that the vast majority of biopores are much smaller than the classical earthworm channel and in fact created by lateral roots (Figure 4; Lucas et al., 2021). Roots were able to create a dense biopore system in a short time period (Figure 3) and developed a dynamic equilibrium in topsoil already after 6 years (approximately 1 vol% and 18 cm cm^{-3}; Lucas et al., 2019b). Lucerne with its large tap root was able to create biopores reaching deep into soil during

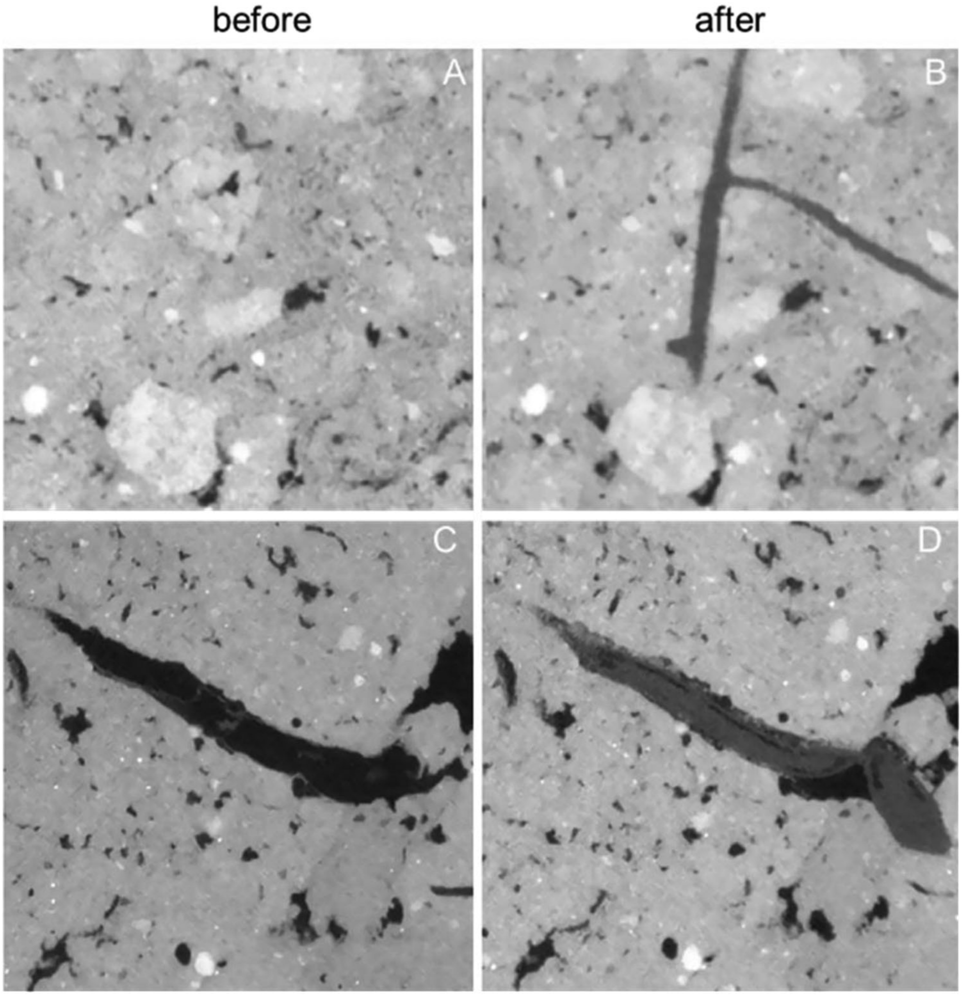

FIGURE 2 Roots elongating into soil, which is sieved (A and B) compared to structured soil (C and D). Shown are slices of ingrowth-cores, scanned at a resolution of 18 µm with µCT before (A and C) an experiment, and after panicum virgatum growing into these (B, D; Lucas et al., unpublished results)

an initial reclamation phase of 3 years. However, the following cereal crops, which could use this biopores to reach deeper soil layers, where the main reason for large biopore densities found in the subsoil. Biopore densities identical to the topsoil were already found after 12 years (Lucas et al., 2019b).

However, there was no change in the PSD of the subsoil over time, as a sufficient volume of macropores such as cracks and packing pores allowed roots to rearrange the exiting macropore structure into a dense system of biopores (Figure 3; Lucas et al., 2019b). Although the creation of this dense biopore system did not affect the PSD, it increased the connectivity of the pore system down to a pore diameter of 0.1 mm due to long ranging connections of lateral roots (Figure 4). This came along with a reduction of the percolation threshold and thus had a positive effect on air permeability and water conductivity near saturation (Lucas et al., 2021).

If the soil does not contain sufficient macropores of the size of roots, they need to grow into the soil matrix. For the pore size distribution, this means that macropores are created in the expense of smaller macro- or mesopores, as this leads to a compaction of the surrounding of the root. This situation typically occurs if the macroporous system was destroyed by compaction (Colombi & Keller, 2019). Whether roots are able to do so, however, mainly depends on the penetration resistance of the soil (Bengough et al., 2011). In general, soil penetration resistance, root architecture and water uptake are closely related (Colombi et al., 2018). Root elongation rate decreases in response to increasing penetrometer resistance and decreasing matric potential in a nonlinear way while the highest decrease of root elongation occurs already under relative wet conditions (Bengough et al., 2011; Veen & Boone, 1990). Thus, root elongation is often reduced even in relatively moist soils (Bengough et al., 2011). Lucas et al. (2019a) investigated how maize roots grow into sieved soil with X-ray µCT, depending on its bulk density. While most roots were able to grow into macropores at a bulk density of 1.3 g cm^{-3}, at bulk densities of 1.45 g cm^{-3} they mainly grew into the soil matrix by pushing aside soil particles, and thus were defined by compacted rhizospheres. At an even higher bulk density of 1.6 at g cm^{-3}, however, maize roots again only grow into macropores, as they were not able to overcome penetration resistance (Lucas et al., 2019a). In addition, with increasing depth in the soil columns, the

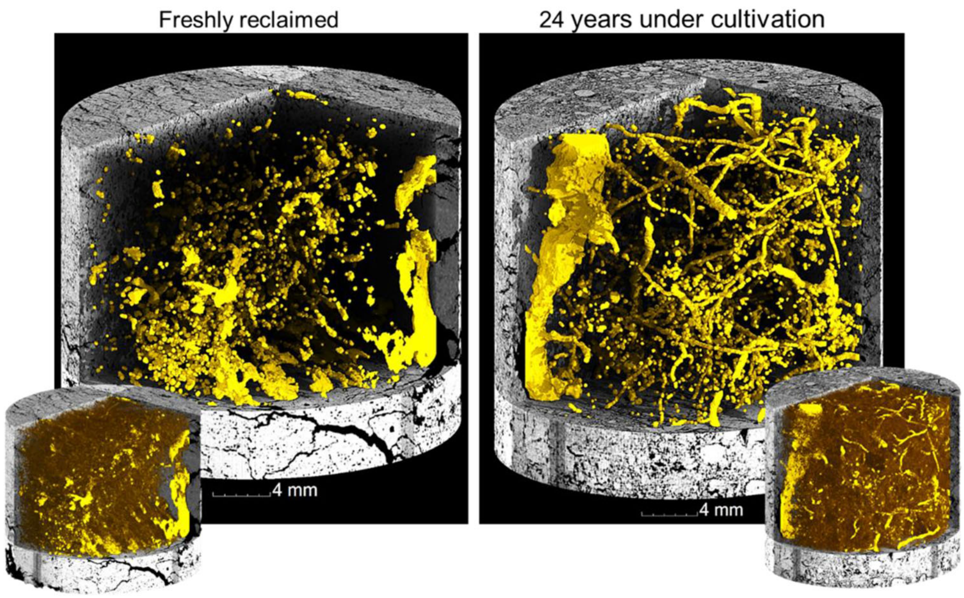

FIGURE 3 Macropores (>0.2 mm) in 3 cm samples after deposit (left) and 24 years after reclamation (right). Small images show all macropores greater 0.019 mm

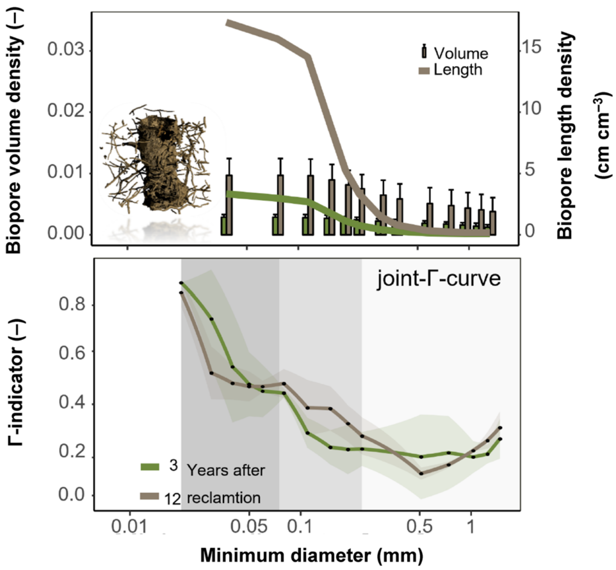

FIGURE 4 The effect of biopore sizes on pore connectivity. Size distribution of biopore volumes and biopore length densities of a subsoil 3 years after recultivation (green) and 12 years after recultivation (brown). Bottom: respective join- Γ-curves as a function of minimum pore diameter. A Γ-value close to 1 implies a perfectly connected system, while Γ-values close to 0 denote for a large number of unconnected pores. Graph from Lucas et al. (2021)

share of roots, which penetrated the dense matrix, decreased resulting in lower root length densities. This suggests that, with lower root length density at higher depth, roots were still able to find sufficient macropores. Thus, the effect of plants on pore structure also highly depends on the ratio of root volume density and macropore volume density (Bodner et al., 2014; Lu et al., 2020; Lucas et al., 2019a). Bodner et al. (2014) measured the effect of 12 different cover crops in soil cores after 4 months of growth on the pore size distribution. These authors showed that the impact of roots on soil porosity seems to be exponential with a lower (0.5 vol%) and upper (1.2 vol%.) limit for root volume density (Bodner et al., 2014).

In general, the concept of the mutual interactions of roots and structure includes, that changes on both sides can lead to better plant root growth. On the one hand, many root traits (amount of laterals, mean root diameter, root volume, etc.) have a potential impact how roots are affected by soil structure and vise-versa (Bodner et al., 2014; Colombi & Keller, 2019; Scholl et al., 2014). Several studies were able to show a positive effect of legumes on macropores especially in the subsoil (Han et al., 2015; Jassogne, 2008; McCallum et al., 2004; Meek et al., 1990, 1992; Mitchell et al., 1995; Perkons et al., 2014; Pulido-Moncada et al., 2020; Uteau et al., 2013), as dicots generally have a higher proportion of thicker roots which are more capable of penetrating dense soil especially due to decreasing buckling or deflecting (Clark et al., 2003; Kautz, 2015; Materechera et al., 1991). This agrees with the findings of Bodner et al. (2014), who showed that coarse root systems with high median root radius increased macroporosity by more than 30%, while decreasing pores <15 µm. The reason for this is that coarse roots mainly need to create new growth paths. Further, species with fine root systems and low median root radius induced heterogenization of the pore space by increasing pores <15 µm. These rooting types were characterized by high root length densities, low penetration strength and thus mainly use the existing growth path (Bodner et al., 2014). On the other hand, improved soil structure through connected biopores can have an important role to overcome limitations of plant growth in compacted soil, as roots can grow into them and thus explore larger soil depths (Ehlers et al., 1983; White & Kirkegaard, 2010). In addition, the importance of macropores for root growth increases with

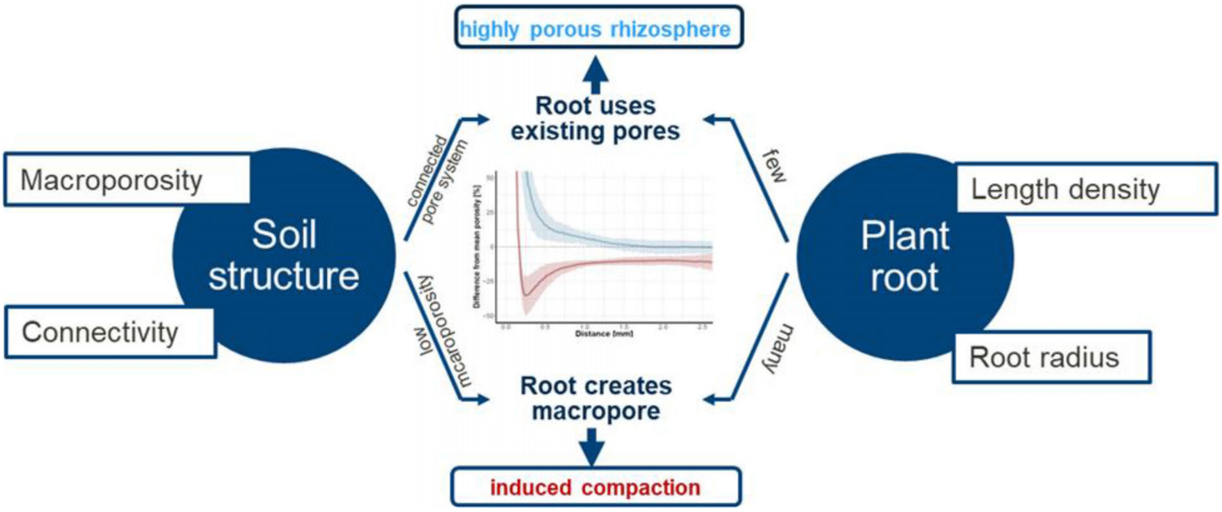

FIGURE 5 Feedback chain of properties of plant roots and soil structure leading to rhizosphere compaction. Graph from Lucas et al. (2019a)

depth, as penetration resistance increases with increasing depth (Gao et al., 2016), and thus even under similar bulk densities, root growth is limited to the use of existing macropores in large depth (White & Kirkegaard, 2010; Zhou et al., 2021). Thus, mean penetrometer resistance seems not to be a good predictor for root growth, as it does not include the effect of macropores. In a recent study, Pandey et al. (2021) questioned the concept of lower root growth as a direct result of increased penetration resistance even more. The authors suggest that in compacted soils, characterized by a low amount of macropores, root growth is actively suppressed by the volatile hormone ethylene, which even induces root morphological changes such as root thickening, typically attributed to root growth in compacted soil. This agrees with findings, that artificial macropores attract root growth toward these especially under high bulk densities (Atkinson et al., 2020; Colombi et al., 2017; Pfeifer et al., 2014). The results of Lucas et al. (2019a) revealed that although maize was barely able to grow into sieved soil at bulk densities of 1.6 g cm^{-3}, under undisturbed field conditions, that is, with a highly connected macropore system, plant roots were still able to reach deep soil layers of even higher bulk density. A high diffusivity of ethylene, which would be associated with macropores (Pandey et al., 2021), may lead to the high share of roots found in biopores in these studies.

6 | HOW THE MUTUAL INTERACTION EFFECTS RHIZOSPHERE PHYSICAL PROPERTIES AND THUS RHIZOSPHERE PROCESSES

As elaborated above, the main driver for difference in rhizosphere physical properties is the mutual interaction of plant roots and pore structure (Figure 5). Thus, if a root grows into a soil with a low amount of unconnected macropores, their rhizosphere will be compacted. In contrast, soils with a high amount of macropores or highly connected biopores lead to a decreased porosity of the rhizosphere, as roots are able to predominately grow into these. Again, the root to macropore ratio seems to be of decisive importance, as only high root length density and volume, under relatively low macropore volumes formed a compacted rhizosphere (Lucas et al., 2019a). Thus, porosity (density) gradients depend in general also on texture and plant species (Helliwell et al., 2019; Phalempin et al., 2021b).

7 | POTENTIAL EFFECTS ON RHIZOSPHERE PROCESSES AND HOW THEY AFFECT OUR EXPERIMENTS

Changes of the rhizosphere physical properties have potential influences on many rhizosphere processes. They directly affect transport of water and nutrients to the root. In model calculations supported by μCT imaging, Aravena et al. (2011) and Aravena et al. (2014) showed that increased rhizosphere compression (reduced porosity) positively affects root water uptake. As discussed above, even compacted rhizospheres feature a more porous space directly at the epidermis, which tremendously affects the ability of roots to take up water and nutrients (Landl et al., 2021).

Root-induced structural changes of the rhizosphere, which come along with a reduction of mean pore size and moisture status around the root, also affect physicochemical reactions in the rhizosphere. In a correlative analysis of synchrotron X-ray fluorescence microscopy and X-ray absorption near-edge structure with X-ray μCT, it was revealed by van Veelen et al. (2020) that the spatial variation in porosity strongly influences iron, sulfur and phosphate chemistry. In detail, dissolution processes were associated with the direct porous vicinity of the root due to optimal conditions for microbial activity, while the subsequent densification zone was defined by reduced permeability and increased binding and reactions on particle surfaces, for example, resulting in the formation of stable Fe oxides (van Veelen et al., 2020).

Changes in the physical protection of organic matter due to differences in the porosity and connectivity of rhizosphere may also change the relative distribution of organic matter in biopore/rhizosphere walls

(Lucas et al., 2020; Rabbi et al., 2016). Recently, it was shown that macropores between 30 and 150 µm favor carbon storage (Kravchenko et al., 2019). Macropores of this size are perfect habitats for microorganisms and by this show increases enzyme activity and carbon turnover (Bouckaert et al., 2013; Kravchenko et al., 2019; Kravchenko et al., 2021; Rabbi et al., 2016). Thus, an overall larger amount of these pore sizes potentially increases carbon storage, as more carbon products can diffuse into the nearby dense soil matrix (Kravchenko et al., 2019). In summary, it can be assumed that a rhizosphere with high porosity close to the epidermis, showing a transition to a dense area (low porosity and small pore sizes), has the potential for increased carbon sequestration.

Likewise, changes in microbial habitats, that is, pore sizes and connectivity are not only changing microbial activity but also microbial community composition (Rabbi et al., 2016; Ruamps et al., 2013). As the macroporosity of the rhizosphere changes, depending on the structures roots encounter (elaborated above), also difference in community composition of the rhizosphere could be partially a result of changing rhizosphere physical properties. Thus, differences in microbial community can be expected when roots grow into loosened, tilled soil compared to no-till soil. Indeed, tillage seems to be one of the main drivers for differences in microbial community composition of the rhizosphere (Babin et al., 2019; Bziuk et al., 2021; Srour et al., 2020).

Further, sieved soil filled into cylinders results in a different macropore system compared to the field even if compacted to the same bulk density (Lucas et al., 2019a). Consequently, different rhizosphere physical properties can be expected. Artifacts created by sieving could be elucidated by including pots with structured soils in our greenhouse experiments.

8 | CONCLUSION AND FUTURE CHALLENGES

Plant roots have the potential to change soil structure and create a connected, dense biopore system over a time. The extent of change, however, is also depending on soil structure itself. In future experiments on the influence of roots on the structure or vice versa, the mutual influence should therefore always be considered. Lucas et al. (2019b) have shown that in a soil with sufficient macroporosity the development of biopore reaches a peak, with astonishing high biopore length densities of approximately 18 cm cm^{-3} already after a period of 6 years. However, how dynamic this threshold is, could not be addressed. Thus, future research should focus on under which circumstances such high biopore densities can be achieved (e.g., soil cultivation and crop rotation) and should also focus on how stable are these biopores created by roots. It remains unclear if biopores smaller than 1 mm are rather reused or destroyed by a root growing close by or how fast they may be filled through feces of mesofauna.

It was elaborated how decisive the mutual interaction of roots and structure can be for the physical properties of the rhizosphere and thus also for major rhizosphere processes and properties. However, little is known about the influences of different root traits on rhizosphere compaction, and in addition, experimental results were often presented as a mean of several or large root segments. Thus, uncertainties remain about the differences of physical properties along a single root or of roots of different size and order. The potential impact on some of the processes affected by changes in rhizosphere physical properties, such as water flux in the rhizosphere, has already been quantified in models (Aravena et al., 2011; Aravena et al., 2014; Landl et al., 2021). Other processes, such as the effect of rhizosphere compaction on carbon storage, are not yet investigated at all. Furthermore, it was shown that through the development of new image analysis pipelines, X-ray µCT has developed to a suitable tool to cover the mutual interaction of roots and structure fully. This enables not just to allow for further understanding of these interactions but also to quantify them. The latter is of particular importance to improve modeling results. Different models could already make use of X-ray µCT imaging results and could implement changes induced by plants on structure or vice versa (Daly et al., 2017; Koebernick et al., 2017; Landl et al., 2019, 2021; Meurer et al., 2020). They can often, however, only draw from a small pool of results. These models can be important not just for upscaling the properties and processes, often imaged at the microscale, they could be also used to combine the effect of different properties and processes to get new insights of their feedback mechanisms (Landl et al., 2021; Meurer et al., 2020; Schnepf et al., 2021).

Lastly, a lot of work in recent years showed that the combination of different imaging techniques with X-ray µCT seems promising to link changes in soil structure to biogeochemical processes (Juyal et al., 2019; Lucas et al., 2020; Schlüter et al., 2018). All these methods, however, have often only been used on a limited number of samples since sample preparation is often time-consuming and tedious. Therefore, it is important to develop effective protocols in the future for both, sample preparation and image analysis, that allow the use of these imaging techniques with high sample numbers. This will minimize uncertainties of results and allow investigating a larger variety of soils and plant species.

ACKNOWLEDGMENTS

I would like to thank my supervisors *Steffen Schluter*, *Hans-Jörg Vogel* and in particular my main supervisor *Doris Vetterlein* for their scientific support, ideas and motivation. Thanks to the *German Research Foundation* (*DFG*) for funding my PhD research within the *research project DFG AOBJ: 628683*. I would like to take this opportunity to thank all collaborators in this project for the interesting and fruitful cooperation. Furthermore, I would like to thank all colleagues at the Department of Soil System Sciences of the UFZ for the very pleasant and inspiring working environment.

Open access funding enabled and organized by Projekt DEAL.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no datasets were generated or analyzed in this study.

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REFERENCES

Amézketa, E. (1999). Soil aggregate stability: A review. *Journal of Sustainable Agriculture*, *14*(2–3), 83–151.

Aravena, J. E., Berli, M., Ghezzehei, T. A., & Tyler, S. W. (2011). Effects of root-induced compaction on rhizosphere hydraulic properties-X-ray microtomography imaging and numerical simulations. *Environmental Science & Technology*, *45*(2), 425–431.

Aravena, J. E., Berli, M., Ruiz, S., Suárez, F., Ghezzehei, T. A., & Tyler, S. W. (2014). Quantifying coupled deformation and water flow in the rhizosphere using X-ray microtomography and numerical simulations. *Plant and Soil*, *376*(1), 95–110.

Athmann, M., Kautz, T., Pude, R., & Köpke, U. (2013). Root growth in biopores—Evaluation with in situ endoscopy. *Plant and Soil*, *371*(1), 179–190.

Atkinson, J. A., Hawkesford, M. J., Whalley, W. R., Zhou, H., & Mooney, S. J. (2020). Soil strength influences wheat root interactions with soil macropores. *Plant, Cell & Environment*, *43*(1), 235–245.

Babin, D., Deubel, A., Jacquiod, S., Sørensen, S. J., Geistlinger, J., Grosch, R., & Smalla, K. (2019). Impact of long-term agricultural management practices on soil prokaryotic communities. *Soil Biology and Biochemistry*, *129*, 17–28.

Bengough, A. G., McKenzie, B. M., Hallett, P. D., & Valentine, T. A. (2011). Root elongation, water stress, and mechanical impedance: A review of limiting stresses and beneficial root tip traits. *Journal of Experimental Botany*, *62*(1), 59–68.

Blaser, S. R., Schlüter, S., & Vetterlein, D. (2018). How much is too much?—Influence of X-ray dose on root growth of faba bean (*Vicia faba*) and barley (*Hordeum vulgare*). *PLoS One*, *13*(3), e0193669. https://doi.org/10.1371/journal.pone.0193669

Bodner, G., Leitner, D., & Kaul, H. P. (2014). Coarse and fine root plants affect pore size distributions differently. *Plant and Soil*, *380*(1), 133–151.

Bouckaert, L., Sleutel, S., Van Loo, D., Brabant, L., Cnudde, V., Van Hoorebeke, L., & De Neve, S. (2013). Carbon mineralisation and pore size classes in undisturbed soil cores. *Soil Research*, *51*(1), 14–22.

Bruand, A., Cousin, I., Nicoullaud, B., Duval, O., & Begon, J. C. (1996). Backscattered electron scanning images of soil porosity for analyzing soil compaction around roots. *Soil Science Society of America Journal*, *60*(3), 895–901.

Bziuk, N., Maccario, L., Douchkov, D., Lueck, S., Babin, D., Sørensen, S. J., Schikora, A., & Smalla, K. (2021). Tillage shapes the soil and rhizosphere microbiome of barley—But not its susceptibility towards *Blumeria graminis* f. sp. *hordei*. *FEMS Microbiology Ecology*, *97*(3), fiab018. https://doi.org/10.1093/femsec/fiab018

Caplan, J., Niethammer, M., Taylor, R. M. II, & Czymmek, K. J. (2011). The power of correlative microscopy: Multi-modal, multi-scale, multi-dimensional. *Current Opinion in Structural Biology*, *21*(5), 686–693.

Carminati, A., Vetterlein, D., Koebernick, N., Blaser, S., Weller, U., & Vogel, H. J. (2013). Do roots mind the gap? *Plant and Soil*, *367*(1), 651–661.

Clark, L. J., Whalley, W. R., & Barraclough, P. B. (2003). How do roots penetrate strong soil? *Plant and Soil*, *255*(1), 93–104.

Colombi, T., Braun, S., Keller, T., & Walter, A. (2017). Artificial macropores attract crop roots and enhance plant productivity on compacted soils. *Science of the Total Environment*, *574*, 1283–1293.

Colombi, T., & Keller, T. (2019). Developing strategies to recover crop productivity after soil compaction—A plant eco-physiological perspective. *Soil and Tillage Research*, *191*, 156–161.

Colombi, T., Torres, L. C., Walter, A., & Keller, T. (2018). Feedbacks between soil penetration resistance, root architecture and water uptake limit water accessibility and crop growth—A vicious circle. *Science of the Total Environment*, *626*, 1026–1035.

Daly, K. R., Cooper, L. J., Koebernick, N., Evaristo, J., Keyes, S. D., Van Veelen, A., & Roose, T. (2017). Modelling water dynamics in the rhizosphere. *Rhizosphere*, *4*, 139–151.

Dexter, A. R. (1987). Compression of soil around roots. *Plant and Soil*, *97*(3), 401–406.

Dıaz-Zorita, M., Perfect, E., & Grove, J. H. (2002). Disruptive methods for assessing soil structure. *Soil and Tillage Research*, *64*(1–2), 3–22.

Downie, H. F., Adu, M. O., Schmidt, S., Otten, W., Dupuy, L. X., White, P. J., & Valentine, T. A. (2015). Challenges and opportunities for quantifying roots and rhizosphere interactions through imaging and image analysis. *Plant, Cell & Environment*, *38*(7), 1213–1232.

Ehlers, W., Köpke, U., Hesse, F., & Böhm, W. (1983). Penetration resistance and root growth of oats in tilled and untilled loess soil. *Soil and Tillage Research*, *3*(3), 261–275.

Feeney, D. S., Crawford, J. W., Daniell, T., Hallett, P. D., Nunan, N., Ritz, K., Rivers, M., & Young, I. M. (2006). Three-dimensional microorganization of the soil–root–microbe system. *Microbial Ecology*, *52*(1), 151–158.

Felde, V. J. M. N. L., Schweizer, S. A., Biesgen, D., Ulbrich, A., Uteau, D., Knief, C., Graf-Rosenfellner, M., Kögel-Knabner, I., & Peth, S. (2021). Wet sieving versus dry crushing: Soil microaggregates reveal different physical structure, bacterial diversity and organic matter composition in a clay gradient. *European Journal of Soil Science*, *72*(2), 810–828.

Flavel, R. J., Guppy, C. N., Rabbi, S. M., & Young, I. M. (2017). An image processing and analysis tool for identifying and analysing complex plant root systems in 3D soil using non-destructive analysis: Root1. *PloS One*, *12*(5), e0176433. https://doi.org/10.1371/journal.pone.0176433

Flavel, R. J., Guppy, C. N., Tighe, M., Watt, M., McNeill, A., & Young, I. M. (2012). Non-destructive quantification of cereal roots in soil using high-resolution X-ray tomography. *Journal of Experimental Botany*, *63*(7), 2503–2511.

Gao, W., Hodgkinson, L., Jin, K., Watts, C. W., Ashton, R. W., Shen, J., Ren, T., Dodd, I. C., Binley, A., Phillips, A. L., Hedden, P., Hawkesford, M. J., & Whalley, W. R. (2016). Deep roots and soil structure. *Plant, Cell & Environment*, *39*(8), 1662–1668.

Gao, W., Schlüter, S., Blaser, S. R., Shen, J., & Vetterlein, D. (2019). A shape-based method for automatic and rapid segmentation of roots in soil from X-ray computed tomography images: Rootine. *Plant and Soil*, *441*(1), 643–655.

Gish, T. J., & Jury, W. A. (1983). Effect of plant roots and root channels on solute transport. *Transactions of the ASAE*, *26*(2), 440–444.

Guyader, J. M., Huizinga, W., Poot, D. H., van Kranenburg, M., Uitterdijk, A., Niessen, W. J., & Klein, S. (2018). Groupwise image registration based on a total correlation dissimilarity measure for quantitative MRI and dynamic imaging data. *Scientific Reports*, *8*(1), 1–14.

Han, E., Kautz, T., Perkons, U., Uteau, D., Peth, S., Huang, N., Horn, R., & Köpke, U. (2015). Root growth dynamics inside and outside of soil biopores as affected by crop sequence determined with the profile wall method. *Biology and Fertility of Soils*, *51*(7), 847–856.

Handschuh, S., Baeumler, N., Schwaha, T., & Ruthensteiner, B. (2013). A correlative approach for combining microCT, light and transmission electron microscopy in a single 3D scenario. *Frontiers in Zoology*, *10*(1), 1–16.

Heeraman, D. A., Hopmans, J. W., & Clausnitzer, V. (1997). Three dimensional imaging of plant roots in situ with X-ray computed tomography. *Plant and Soil*, *189*(2), 167–179.

Helliwell, J. R., Sturrock, C. J., Mairhofer, S., Craigon, J., Ashton, R. W., Miller, A. J., Whalley, W. R., & Mooney, S. J. (2017). The emergent rhizosphere: imaging the development of the porous architecture at the root–soil interface. *Scientific Reports*, *7*(1), 1–10.

Helliwell, J. R., Sturrock, C. J., Miller, A. J., Whalley, W. R., & Mooney, S. J. (2019). The role of plant species and soil condition in the structural development of the rhizosphere. *Plant, Cell & Environment*, *42*(6), 1974–1986.

Hinsinger, P., Bengough, A. G., Vetterlein, D., & Young, I. M. (2009). Rhizosphere: biophysics, biogeochemistry and ecological relevance. *Plant and Soil*, *321*(1), 117–152.

Jarvis, N. J. (2007). A review of non-equilibrium water flow and solute transport in soil macropores: Principles, controlling factors and consequences for water quality. *European Journal of Soil Science*, *58*(3), 523–546.

Jassogne, L. (2008). *Characterisation of porosity and root growth in a sodic texture-contrast soil*. (Doctoral dissertation, University of Western Australia).

Jin, K., Shen, J., Ashton, R. W., Dodd, I. C., Parry, M. A., & Whalley, W. R. (2013). How do roots elongate in a structured soil? *Journal of Experimental Botany*, *64*(15), 4761–4777.

Jin, K., White, P. J., Whalley, W. R., Shen, J., & Shi, L. (2017). Shaping an optimal soil by root–soil interaction. *Trends in Plant Science*, *22*(10), 823–829.

Juyal, A., Otten, W., Falconer, R., Hapca, S., Schmidt, H., Baveye, P. C., & Eickhorst, T. (2019). Combination of techniques to quantify the distribution of bacteria in their soil microhabitats at different spatial scales. *Geoderma*, *334*, 165–174.

Kautz, T. (2015). Research on subsoil biopores and their functions in organically managed soils: A review. *Renewable Agriculture and Food Systems*, *30*(4), 318–327.

Kautz, T., Perkons, U., Athmann, M., Pude, R., & Köpke, U. (2013). Barley roots are not constrained to large-sized biopores in the subsoil of a deep Haplic Luvisol. *Biology and Fertility of Soils*, *49*(7), 959–963.

Koebernick, N., Daly, K. R., Keyes, S. D., Bengough, A. G., Brown, L. K., Cooper, L. J., George, T. S., Hallett, P. D., Naveed, M., Raffan, A., & Roose, T. (2019). Imaging microstructure of the barley rhizosphere: Particle packing and root hair influences. *New Phytologist*, *221*(4), 1878–1889.

Koebernick, N., Daly, K. R., Keyes, S. D., George, T. S., Brown, L. K., Raffan, A., Cooper, L. J., Naveed, M., Bengough, A. G., Sinclair, I., Hallett, P. D., & Roose, T. (2017). High-resolution synchrotron imaging shows that root hairs influence rhizosphere soil structure formation. *New Phytologist*, *216*(1), 124–135.

Koebernick, N., Weller, U., Huber, K., Schlüter, S., Vogel, H.-J., Jahn, R., Vereecken, H., & Vetterlein, D. (2014). In situ visualization and quantification of three-dimensional root system architecture and growth using X-ray computed tomography. *Vadose Zone Journal*, *13*(8), vzj2014–03.

Koestel, J., Dathe, A., Skaggs, T. H., Klakegg, O., Ahmad, M. A., Babko, M., Giménez, D., Farkas, C., Nemes, A., & Jarvis, N. (2018). Estimating the permeability of naturally structured soil from percolation theory and pore space characteristics imaged by X-ray. *Water Resources Research*, *54*(11), 9255–9263.

Koestel, J., & Larsbo, M. (2014). Imaging and quantification of preferential solute transport in soil macropores. *Water Resources Research*, *50*(5), 4357–4378.

Kravchenko, A., & Guber, A. (2021). Imaging soil structure to measure soil functions and soil health with X-ray computed micro-tomography. In W. Otten (Ed.), *Advances in measuring soil health* (pp. 111–138). Burleigh Dodds Science Publishing,.

Kravchenko, A., Guber, A., Gunina, A., Dippold, M., & Kuzyakov, Y. (2021). Pore-scale view of microbial turnover: Combining ^{14}C imaging, µCT and zymography after adding soluble carbon to soil pores of specific sizes. *European Journal of Soil Science*, *72*(2), 593–607.

Kravchenko, A., Guber, A. K., Razavi, B. S., Koestel, J., Quigley, M. Y., Robertson, G. P., & Kuzyakov, Y. (2019). Microbial spatial footprint as a driver of soil carbon stabilization. *Nature Communications*, *10*(1), 1–10.

Landl, M., Phalempin, M., Schlüter, S., Vetterlein, D., Vanderborght, J., Kroener, E., & Schnepf, A. (2021). Modeling the impact of rhizosphere bulk density and mucilage gradients on root water uptake. *Frontiers in Agronomy*, *3*, 6. https://doi.org/10.3389/fagro.2021.622367

Landl, M., Schnepf, A., Uteau, D., Peth, S., Athmann, M., Kautz, T., Perkons, U., Vereecken, H., & Vanderborght, J. (2019). Modeling the impact of biopores on root growth and root water uptake. *Vadose Zone Journal*, *18*(1), 1–20.

Lin, H. (2010). Earth's critical zone and hydropedology: Concepts, characteristics, and advances. *Hydrology and Earth System Sciences*, *14*(1), 25–45.

Lontoc-Roy, M., Dutilleul, P., Prasher, S. O., Han, L., Brouillet, T., & Smith, D. L. (2006). Advances in the acquisition and analysis of CT scan data to isolate a crop root system from the soil medium and quantify root system complexity in 3-D space. *Geoderma*, *137*(1–2), 231–241.

Lontoc-Roy, M., Dutilleul, P., Prasher, S. O., Han, L., & Smith, D. L. (2005). Computed tomography scanning for three-dimensional imaging and complexity analysis of developing root systems. *Botany*, *83*(11), 1434–1442.

Lu, J., Zhang, Q., Werner, A. D., Li, Y., Jiang, S., & Tan, Z. (2020). Root-induced changes of soil hydraulic properties—A review. *Journal of Hydrology*, *589*, 125203. https://doi.org/10.1016/j.jhydrol.2020.125203

Lucas, M., Pihlap, E., Steffens, M., Vetterlein, D., & Kögel-Knabner, I. (2020). Combination of imaging infrared spectroscopy and X-ray computed microtomography for the investigation of bio-and physicochemical processes in structured soils. *Frontiers in Environmental Science*, *8*, 42. https://doi.org/10.3389/fenvs.2020.00042

Lucas, M., Schlüter, S., Vogel, H. J., & Vetterlein, D. (2019a). Roots compact the surrounding soil depending on the structures they encounter. *Scientific Reports*, *9*(1), 1–13.

Lucas, M., Schlüter, S., Vogel, H. J., & Vetterlein, D. (2019b). Soil structure formation along an agricultural chronosequence. *Geoderma*, *350*, 61–72.

Lucas, M., Vetterlein, D., Vogel, H. J., & Schlüter, S. (2021). Revealing pore connectivity across scales and resolutions with X-ray CT. *European Journal of Soil Science*, *72*(2), 546–560.

Mairhofer, S., Zappala, S., Tracy, S. R., Sturrock, C., Bennett, M., Mooney, S. J., & Pridmore, T. (2012). RooTrak: Automated recovery of three-dimensional plant root architecture in soil from X-ray microcomputed tomography images using visual tracking. *Plant Physiology*, *158*(2), 561–569.

Materechera, S. A., Dexter, A. R., & Alston, A. M. (1991). Penetration of very strong soils by seedling roots of different plant species. *Plant and Soil*, *135*(1), 31–41.

McCallum, M. H., Kirkegaard, J. A., Green, T. W., Cresswell, H. P., Davies, S. L., Angus, J. F., & Peoples, M. B. (2004). Improved subsoil macroporosity following perennial pastures. *Australian Journal of Experimental Agriculture*, *44*(3), 299–307.

Meek, B. D., DeTar, W. R., Rolph, D., Rechel, E. R., & Carter, L. M. (1990). Infiltration rate as affected by an alfalfa and no-till cotton cropping system. *Soil Science Society of America Journal*, *54*, 505–508.

Meek, B. D., Rechel, E. R., Carter, L. M., DeTar, W. R., & Urie, A. L. (1992). Infiltration rate of a sandy loam soil: Effects of traffic, tillage, and plant roots. *Soil Science Society of America Journal*, *56*(3), 908–913.

Meurer, K., Barron, J., Chenu, C., Coucheney, E., Fielding, M., Hallett, P., Hermann, A. M., Keller, T., Koestel, J., Larsbo, M., Lewan, E., Or, D., Parsons, D., Parvin, N., Taylor, A., Vereecken, H., & Jarvis, N. (2020). A framework for modelling soil structure dynamics induced by biological activity. *Global Change Biology*, *26*(10), 5382–5403.

Mitchell, A. R., Ellsworth, T. R., & Meek, B. D. (1995). Effect of root systems on preferential flow in swelling soil. *Communications in Soil Science and Plant Analysis*, *26*(15–16), 2655–2666.

Naveed, M., Moldrup, P., Arthur, E., Wildenschild, D., Eden, M., Lamandé, M., Vogel, H.-J., & De Jonge, L. W. (2013). Revealing soil structure and functional macroporosity along a clay gradient using X-ray computed tomography. *Soil Science Society of America Journal*, *77*(2), 403–411.

Nimmo, J. R. (2013). *Porosity and pore size distribution, reference module in earth systems and environmental sciences*. Elsevier.

Pandey, B. K., Huang, G., Bhosale, R., Hartman, S., Sturrock, C. J., Jose, L., Martin, O. C., Karady, M., Voesenek, L. A. C. J., Ljung, K., Lynch, J. P., Brown, K. M., Whalley, W. R., Mooney, S. J., Zhang, D., & Bennett, M. J. (2021). Plant roots sense soil compaction through restricted ethylene diffusion. *Science*, *371*(6526), 276–280.

Perkons, U., Kautz, T., Uteau, D., Peth, S., Geier, V., Thomas, K., Lütke Holz, K., Athmann, M., Pude, R., & Köpke, U. (2014). Root-length densities of various annual crops following crops with contrasting root systems. *Soil and Tillage Research*, *137*, 50–57.

Pfeifer, J., Kirchgessner, N., & Walter, A. (2014). Artificial pores attract barley roots and can reduce artifacts of pot experiments. *Journal of Plant Nutrition and Soil Science*, *177*(6), 903–913.

Phalempin, M., Lippold, E., Vetterlein, D., & Schlüter, S. (2021a). An improved method for the segmentation of roots from X-ray computed tomography 3D images: Rootine v. 2. *Plant Methods*, *17*(1), 1–19.
Phalempin, M., Lippold, E., Vetterlein, D., & Schlüter, S. (2021b). Soil texture and structure heterogeneity predominantly governs bulk density gradients around roots. *Vadose Zone Journal*, *20*(5), e20147.
Pierret, A., Capowiez, Y., Moran, C. J., & Kretzschmar, A. (1999). X-ray computed tomography to quantify tree rooting spatial distributions. *Geoderma*, *90*(3–4), 307–326.
Pierret, A., Hartmann, C., Maeght, J. L., Pagès, L. (2011). Biotic regulation: Plants. In K. Ritz, & I. Young (Eds.), *The architecture and biology of soils. Life in inner space*. CABI.
Pihlap, E., Vuko, M., Lucas, M., Steffens, M., Schloter, M., Vetterlein, D., Endenich, M., & Kögel-Knabner, I. (2019). Initial soil formation in an agriculturally reclaimed open-cast mining area-the role of management and loess parent material. *Soil and Tillage Research*, *191*, 224–237.
Pulido-Moncada, M., Katuwal, S., Ren, L., Cornelis, W., & Munkholm, L. (2020). Impact of potential bio-subsoilers on pore network of a severely compacted subsoil. *Geoderma*, *363*, 114154. https://doi.org/10.1016/j.geoderma.2019.114154
Rabbi, S. M. F., Daniel, H., Lockwood, P. V., Macdonald, C., Pereg, L., Tighe, M., Wilson, B. R., & Young, I. M. (2016). Physical soil architectural traits are functionally linked to carbon decomposition and bacterial diversity. *Scientific Reports*, *6*(1), 1–9.
Rabbi, S. M. F., Tighe, M. K., Flavel, R. J., Kaiser, B. N., Guppy, C. N., Zhang, X., & Young, I. M. (2018). Plant roots redesign the rhizosphere to alter the three-dimensional physical architecture and water dynamics. *New Phytologist*, *219*(2), 542–550.
Rabbi, S. M. F., Tighe, M. K., Knox, O., & Young, I. M. (2018). The impact of carbon addition on the organisation of rhizosheath of chickpea. *Scientific Reports*, *8*(1), 1–9.
Rabot, E., Wiesmeier, M., Schlüter, S., & Vogel, H. J. (2018). Soil structure as an indicator of soil functions: A review. *Geoderma*, *314*, 122–137.
Rasse, D. P., Smucker, A. J., & Santos, D. (2000). Alfalfa root and shoot mulching effects on soil hydraulic properties and aggregation. *Soil Science Society of America Journal*, *64*(2), 725–731.
Renard, P., & Allard, D. (2013). Connectivity metrics for subsurface flow and transport. *Advances in Water Resources*, *51*, 168–196.
Ruamps, L. S., Nunan, N., Pouteau, V., Leloup, J., Raynaud, X., Roy, V., & Chenu, C. (2013). Regulation of soil organic C mineralisation at the pore scale. *FEMS Microbiology Ecology*, *86*(1), 26–35.
Scanlan, C. A. (2009). *Processes and effects of root-induced changes to soil hydraulic properties. (Doctoral dissertation*, University of Western Australia).
Schindelin, J., Arganda-Carreras, I., Frise, E., Kaynig, V., Longair, M., Pietzsch, T., Preibisch, S., Rueden, C., Saalfeld, S., Schmid, B., Tinevez, J.-Y., White, D. J., Hartenstein, V., Eliceiri, K., Tomancak, P., & Cardona, A. (2012). Fiji: An open-source platform for biological-image analysis. *Nature Methods*, *9*(7), 676–682.
Schlüter, S., Eickhorst, T., & Mueller, C. W. (2018). Correlative imaging reveals holistic view of soil microenvironments. *Environmental Science & Technology*, *53*(2), 829–837.
Schnepf, A., Carminati, A., Ahmed, M. A., Ani, M., Benard, P., Bentz, J., Bonkowski, M., Brax, M., Diehl, D., Duddek, P., Kröner, E., Javaux, M., Landl, M., Lehndorff, E., Lippold, E., Lieu, A., Mueller, C. W., Oburger, E., Otten, W., … Vetterlein, D. (2021). Linking rhizosphere processes across scales: Opinion. *bioRxiv*. https://doi.org/10.1101/2021.07.08.451655
Scholl, P., Leitner, D., Kammerer, G., Loiskandl, W., Kaul, H. P., & Bodner, G. (2014). Root induced changes of effective 1D hydraulic properties in a soil column. *Plant and Soil*, *381*(1), 193–213.
Schweizer, S. A., Bucka, F. B., Graf-Rosenfellner, M., & Kögel-Knabner, I. (2019). Soil microaggregate size composition and organic matter distribution as affected by clay content. *Geoderma*, *355*, 113901. https://doi.org/10.1016/j.geoderma.2019.113901
Srour, A. Y., Ammar, H. A., Subedi, A., Pimentel, M., Cook, R. L., Bond, J., & Fakhoury, A. M. (2020). Microbial communities associated with long-term tillage and fertility treatments in a corn–soybean cropping system. *Frontiers in Microbiology*, *11*, 1363. https://doi.org/10.3389/fmicb.2020.01363
Stirzaker, R. J., Passioura, J. B., & Wilms, Y. (1996). Soil structure and plant growth: Impact of bulk density and biopores. *Plant and Soil*, *185*(1), 151–162.
Tippkötter, R. (1983). Morphology, spatial arrangement and origin of macropores in some hapludalfs, West Germany. *Geoderma*, *29*(4), 355–371.
Tracy, S. R., Black, C. R., Roberts, J. A., & Mooney, S. J. (2013). Exploring the interacting effect of soil texture and bulk density on root system development in tomato (*Solanum lycopersicum* L.). *Environmental and Experimental Botany*, *91*, 38–47.
Uteau, D., Pagenkemper, S. K., Peth, S., & Horn, R. (2013). Root and time dependent soil structure formation and its influence on gas transport in the subsoil. *Soil and Tillage Research*, *132*, 69–76.
Veelen, A., Koebernick, N., Scotson, C. S., Mckay-Fletcher, D., Huthwelker, T., Borca, C. N., Mosselmans, J. F. W., & Roose, T. (2020). Root-induced soil deformation influences Fe, S and P: rhizosphere chemistry investigated using synchrotron XRF and XANES. *New Phytologist*, *225*(4), 1476–1490.
Veen, B. W., & Boone, F. R. (1990). The influence of mechanical resistance and soil water on the growth of seminal roots of maize. *Soil and Tillage Research*, *16*(1–2), 219–226.
Vetterlein, D., Carminati, A., Kögel-Knabner, I., Bienert, G. P., Smalla, K., Oburger, E., Schnepf, A., Banitz, T., Tarkka, M. T., & Schlüter, S. (2020). Rhizosphere spatiotemporal organization–A key to Rhizosphere functions. *Frontiers in Agronomy*, *2*, 8. https://doi.org/10.3389/fagro.2020.00008
Vogel, H. J., Balseiro-Romero, M., Kravchenko, A., Otten, W., Pot, V., Schlüter, S., Weller, U., & Baveye, P. C. (2021). A holistic perspective on soil architecture is needed as a key to soil functions. *European Journal of Soil Science*. https://doi.org/10.1111/ejss.13152
Vogel, H. J., Weller, U., & Schlüter, S. (2010). Quantification of soil structure based on Minkowski functions. *Computers & Geosciences*, *36*(10), 1236–1245.
Vollsnes, A. V., Futsaether, C. M., & Bengough, A. G. (2010). Quantifying rhizosphere particle movement around mutant maize roots using time-lapse imaging and particle image velocimetry. *European Journal of Soil Science*, *61*(6), 926–939.
White, R. G., & Kirkegaard, J. A. (2010). The distribution and abundance of wheat roots in a dense, structured subsoil–implications for water uptake. *Plant, Cell & Environment*, *33*(2), 133–148.
Wuest, S. B. (2001). Soil biopore estimation: effects of tillage, nitrogen, and photographic resolution. *Soil and Tillage Research*, *62*(3–4), 111–116.
Young, I. M. (1998). Biophysical interactions at the root–soil interface: a review. *The Journal of Agricultural Science*, *130*(1), 1–7.
Yunusa, I. A., & Newton, P. J. (2003). Plants for amelioration of subsoil constraints and hydrological control: The primer-plant concept. *Plant and Soil*, *257*(2), 261–281.
Zhou, Hu, Whalley, W. R., Hawkesford, M. J., Ashton, R. W., Atkinson, B., Atkinson, J. A., Sturrock, C. J., Bennett, M. J., & Mooney, S. J. (2021). The interaction between wheat roots and soil pores in structured field soil. *Journal of Experimental Botany*, *72*(2), 747–756.

How to cite this article: Lucas, M. (2022). Perspectives from the Fritz-Scheffer Awardee 2020—The mutual interactions between roots and soil structure and how these affect rhizosphere processes. *Journal of Plant Nutrition and Soil Science*, *185*, 8–18. https://doi.org/10.1002/jpln.202100385